

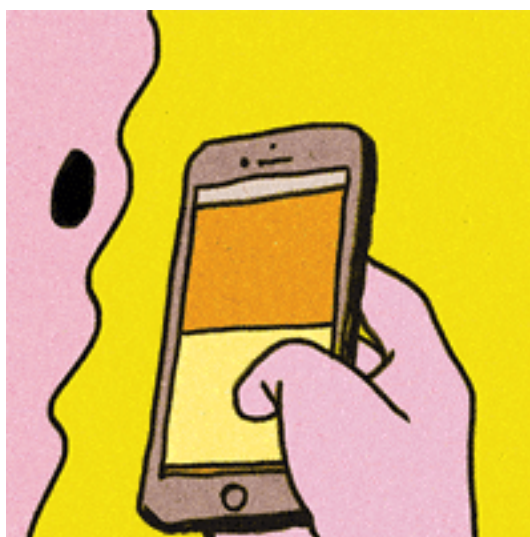
ECS172 Group Project Description

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Department of Computer Science, UC Davis



Why is this additive?

Overview

The final group project is a major component of ECS172, accounting for **50% of your final grade**. This project gives you the opportunity to apply the tools and concepts learned throughout the quarter to a challenging, real-world recommendation system problem using machine learning.

The goal of the final project is to identify a novel directions of improving recommendation systems, which can be but not limited to the following directions:

- Improving an existing recommendation system algorithm.
- Identify novel ways recommendation systems can be used.
- Better integration of LLMs with recommendation systems.

The goal of the final project is **not only to apply learned models and knowledge**, but also to **extend or improve them in the context of novel techniques or novel problems**. This could

mean identifying an important but underexplored problem in recommendation systems, or adapting and enhancing existing algorithms to perform better in a specific problem.

Projects should be driven by practical relevance, solid motivation, and technical rigor. You will work in teams of **three students** to explore the recommendation system techniques and research—from problem definition and literature review, to proposed approach, experiment design, evaluation, and final presentation.

Project Team Guidelines

- Students form groups of three members. (**register team members** [HERE](#))
- Team composition must be finalized by the end of Week 4.
- All submissions (proposal, presentation, paper) must be prepared and submitted as a team.

Project Requirements

Your project must involve the design and implementation of a recommendation system task. Your team may focus on one or both of the following novelty axes:

1. **Problem Novelty:** Identify a new and impactful real-world problem that has not been well explored in the Recsys and ML literature, and demonstrate how important it problem potentially is and can be solved with your proposed approach.
2. **Algorithmic Novelty:** Propose a novel and well-motivated Recsys/ML algorithms or methodologies that are better suited for the chosen Recsys task than existing methods.

All projects must:

- Use real-world data, or public online datasets. Only using simulated or synthetic datasets are not allowed unless used in addition to a real dataset for analysis.
- Define a clear problem statement and objective (e.g., retrieval/ranking of XXX, filter bubble, bias correction, etc.).
- To justify novelty: use at least three papers published in the following venues in the past three years
 - Main conferences of Recsys, WWW, KDD, ICML, ICLR, NeurIPS, etc.
 - Journals including TACL, JMLR, TPAMI, NLP/CV conferences, such as EMNLP, ACL, ICCV, CVPR
 - No other venues or workshop papers are allowed
- Description of dataset(s) and evaluation protocol (incl. metrics)
- Clearly state your evaluation protocol and metrics (e.g., accuracy, RMSE, NDCG, human evaluation, LLM as Judge).
- Discuss limitations, ethical considerations, and challenges encountered.

Use of Pretrained Models and Resources

You may use pre-trained large models (e.g., BERT, ResNet, CLIP), provided they are accessible via public APIs or platforms such as Hugging Face or TensorFlow Hub.

Important: The course does not provide computing resources (e.g., GPUs or cloud credits). Your project must be computationally feasible using resources you can access (e.g., Google Colab).

Grading Breakdown (50% of Final Grade)

Component	Weight
Project Proposal	10%
Project Presentation	15%
Final Paper	25%

Project Proposal (10%)

Due Date: End of Week 5

Length: 4 pages maximum (excluding references)

The proposal should describe:

- The project's motivation and goals
- A detailed problem statement
- Justification of novelty using relevant literature
- Proposed approach or methodology
- Dataset(s) and data acquisition methods
- Evaluation metrics and plans

Project Presentation (15%)

Due Date: End of week 9 (Slides)

Presentation Window: Weeks 10 during scheduled class times

Each team will deliver a 6 minute presentation summarizing the key components of their project.

Final Paper (25%)

Due Date: Finals Week, end of week 10

Length: Up to 8 pages, excluding references.

Format: Conference-style full paper format. [Template available at ACM \(section 2 workflow and template\)](#)

Your final paper must be professionally written and structured, An example organization of the paper will contain:

- Abstract (required)
- Introduction (required)
- Related Work (required)
- Methodology
- Experiments and Results
- Discussion and Analysis
- Conclusion (required)
- **Contribution Statement (required):**
A sentence for each team member of their contributions.
- References (required)

Note: If you have questions at any stage of the project, please contact instructor or TAs, or reach out during office hours or on the course discussion forum. Early feedback is encouraged.